PDS Search LDD Form

Scientific Discipline

Information about the scientific discipline and research context.

Primary Field What are the different scientific fields of study.

- [ ] **Astrobiology** Study of life in the universe
- [ ] **Astronomy Observation** Observation of celestial objects
- [ ] **Astronomy Radar** Radar observations of celestial objects
- [ ] **Astronomy Exoplanets** Study of exoplanets
- [ ] **Astronomy Spectroscopy** Spectroscopy of celestial objects
- [ ] **Geology Geophysics** Geophysics of the Earth and other planets
- [ ] **Geology Seismology** Seismology of the Earth and other planets
- [ ] **Geology Surface** Surface studies of the Earth and other planets
- [ ] **Geology Crater Counting** Crater counting of the Earth and other planets
- [ ] **Geology Interior Studies** Interior studies of the Earth and other planets
- [ ] **Dynamics** Dynamics of the Earth and other planets
- [ ] **Laboratory Geochemistry** Geochemistry of physical samples
- [ ] **Laboratory Cosmochemistry** Cosmochemistry of physical samples
- [ ] **Laboratory Age Dating** Age dating of physical samples
- [ ] **Laboratory Analytic Chemistry** Analytic chemistry of physical samples
- [ ] **Laboratory Spectral Analysis** Spectral analysis of physical samples
- [ ] **Atmospheric Study** Atmospheric studies of the Earth and other planets
- [ ] **Impact** Measuring and categorizing impact craters and rates
- [ ] **Hydrology** Fluvial mechanics
- [ ] **Exoplanets** Detection and characterizing exoplanets
- [ ] **Modeling** Modeling of the Earth and other planets
- [ ] **Magnetospheres** Magnetospheres of the Earth and other planets
- [ ] **Space Physics** Space physics of the Earth and other planets
- [ ] **Other** Any field not covered by other values

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Investigation Technique What method of scientific investigations are used.

- [ ] **Spectroscopy** Collection of spectra
- [ ] **Modeling** Computer-based simulation or modeling
- [ ] **Mapping** Collection of maps
- [ ] **Radar** Collection of radar data
- [ ] **Seismology** Collection of seismic data
- [ ] **Ground based observation** Data that was derived from Earth-based telescopes
- [ ] **Gravimetry** Collection of gravity data
- [ ] **Remove Sensing** Various data collected via remote sensing techinques
- [ ] **Geochemistry** Collection of geochemical/cosmochmical data
- [ ] **Laboratory** Various laboratory work
- [ ] **Artificial Intelligence** AI and machine learning techniques
- [ ] **ICPMS** Collection of inductively coupled plasma mass spectrometry data
- [ ] **Other** Any investigation technique not covered by other values

Study Focus Specific area of study within the scientific discipline.

- ☐ **Surface** Study focused on planetary geology such as features or properties
- ☐ **Interior** Study focused on internal structure or processes
- ☐ **Atmosphere** Study focused on atmospheric properties or dynamics
- ☐ **Magnetosphere** Study focused on magnetic field properties or interactions
- ☐ **Orbital Dynamics** Study focused on orbital properties or behaviors
- ☐ **Composition** Study focused on chemical or mineral composition
- ☐ **Plasma** Study of plasma and ionospheres
- ☐ **Space Weather** Study of the change in surfaces due to space weather effects
- ☐ **Planetary Protection** Study focused on planetary protection
- ☐ **Planetary Rings** Study focused on ring and ring-moon systems
- ☐ **Planetary Resources** In-situ resource characterization and utilization
- ☐ **Other** Any focus not covered by other values

Target Object

Information about the target of observation, whether it is a celestial body, feature, or other object.

Object Name Name of the object

Value: ______________________

Parent Body Parent body of the object

Value: ______________________

Scope Scope of the data collection

- ☐ **Global** Data collected from the entire object
- ☐ **Regional** Data collected from a large region of the object
- ☐ **Local** Data collected from a small location on the object
- ☐ **Other** Any scope not covered by other values

Object Type

Select the categories within which the data would fit into

Planet Planet

- ☐ **Gas Giant** Large planets composed mainly of gas
- ☐ **Terrestrial** Rocky planets similar to Earth
- ☐ **Interior** Planets closer to the Sun than Earth
- ☐ **Other** Any planet type not covered by other values

Particles and Fields Fields of an object

- ☐ **Particles** Particles in the atmosphere
- ☐ **Cosmic Rays** Cosmic rays in the atmosphere
- ☐ **X-Rays** X-rays in the atmosphere
- ☐ **Gamma-Rays** Gamma-rays in the atmosphere
- ☐ **Electromagnetic Waves** Electromagnetic waves in the atmosphere

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Dust Dust

- ☐ **Meteoroids** Dust from meteoroids
- ☐ **IPD** Interplanetary Dust
- ☐ **Comet** Dust from comets
- ☐ **Other** Other dust

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Solar Component Solar component

- ☐ **Corona** Corona
- ☐ **Solar Wind** Solar Wind
- ☐ **Exoplanets** Exoplanets
- ☐ **Heliosphere** Heliosphere
- ☐ **Core** Core
- ☐ **Radiative Zone** Radiative Zone
- ☐ **Convective Zone** Convective Zone
- ☐ **Photosphere** Photosphere
- ☐ **Chromosphere** Chromosphere
- ☐ **Other** Any solar component not covered by other values

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Exoplanet Type of an exoplanet

- ☐ **Terrestrial** Rocky planets, up to 1.5 Earth radii
- ☐ **Gas Giants** Larger than 10 Earth radii
- ☐ **Ice Giants** 4-6 Earth radii
- ☐ **Super-Earths** 1.5 to 2 Earth radii
- ☐ **Mini-Neptunes** 2-04 Earth radii
- ☐ **Lava Worlds** Molten rock
- ☐ **Ocean Worlds** Global oceans
- ☐ **Rogue Planets** Planets not bound to a star
- ☐ **Other** Other categories

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Small Body Small body

Asteroid Asteroid

- [ ] **Near Earth** Asteroids with orbits that bring them close to Earth
- [ ] **Trojan** Asteroids that share an orbit with a planet
- [ ] **Active** Asteroids that show comet-like activity
- [ ] **Main Belt** Asteroids in the main asteroid belt
- [ ] **PHA** Potential Hazardous Asteroid
- [ ] **Other** Any asteroid type not covered by other values

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TNO Trans-Neptunian Object

- [ ] **Classical** Objects in the classical Kuiper belt
- [ ] **Resonant** Objects in orbital resonance with Neptune
- [ ] **Scattered** Objects scattered by Neptune
- [ ] **Detached** Objects with orbits not strongly influenced by Neptune
- [ ] **Other** Any TNO type not covered by other values

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Centaur Centaur

- [ ] **Active** Centaurs showing comet-like activity
- [ ] **Inactive** Centaurs without comet-like activity
- [ ] **Other** Any centaur type not covered by other values

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KBO Kuiper Belt Object

- [ ] **Classical** Objects in the classical Kuiper belt
- [ ] **Resonant** Objects in orbital resonance with Neptune
- [ ] **Scattered** Objects scattered by Neptune
- [ ] **Other** Any KBO type not covered by other values

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Oort Cloud Oort Cloud Object

- [ ] **Inner** Objects in the inner Oort cloud
- [ ] **Outer** Objects in the outer Oort cloud
- [ ] **Other** Any Oort cloud type not covered by other values

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Comet Comet

- [ ] **Jupiter Family** Comets that orbit Jupiter
- [ ] **Single Pass** Comets that pass through the inner solar system
- [ ] **Other** Any comet type not covered by other values

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Satellite Satellite

- [ ] **Icy Moons** Icy moons of a planet
- [ ] **Galilean Moons** Galilean moons of Jupiter
- [ ] **Irregular Moons** Irregular moons of a planet
- [ ] **Captured Moons** Captured moons of a planet
- [ ] **Shepherding Moon** Shepherding moon of a planet
- [ ] **Other** Any satellite type not covered by other values

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Feature Feature

Name Name of the feature

Value: ______________________

Type Type of the feature

- [ ] **Crater** Impact depression on a surface
- [ ] **Mountain** Elevated landform
- [ ] **Valley** Elongated depression
- [ ] **Mare** Dark plains on the Moon
- [ ] **Volcano** Volcanic landform
- [ ] **Cloud** Atmospheric cloud formation
- [ ] **Other** Any feature type not covered by other values

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Atmosphere Atmosphere of a celestial body

Layer Layer of the atmosphere

- [ ] **Troposphere** Lowest layer of the atmosphere
- [ ] **Stratosphere** Layer above the troposphere
- [ ] **Ionosphere** Layer containing ions
- [ ] **Exosphere** Outermost layer of the atmosphere
- [ ] **Other** Any atmospheric layer not covered by other values

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Study Type Type of study of the atmosphere

- [ ] **Temperature Profile** Temperature profile of the atmosphere
- [ ] **Scale Height** Scale height of the atmosphere
- [ ] **Density Profile** Density profile of the atmosphere

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Rings Rings of an object

Ring Type Type of ring

- [ ] **D** D ring
- [ ] **C** C ring
- [ ] **B** B ring
- [ ] **A** A ring
- [ ] **F** F ring
- [ ] **G** G ring
- [ ] **E** E ring

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Ring Division Division of the ring

- [ ] **Cassini Division** Cassini Division
- [ ] **Encke** Encke Division
- [ ] **Keeler** Keeler Division

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Ring Feature Feature of the ring

- [ ] **Spiral Bending Waves** Spiral Bending Waves
- [ ] **Spokes** Spokes
- [ ] **Spiral Density Waves** Spiral Density Waves

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Physical Properties

Information about the physical properties of the object under study.

Material Type Material type of the object

- [ ] **Rock** Solid mineral material
- [ ] **Ice** Frozen water or other volatiles
- [ ] **Gas** Gaseous material
- [ ] **Dust** Fine particulate material
- [ ] **Regolith** Surface layer of loose material
- [ ] **Liquid** Liquid state material
- [ ] **Other** Any material type not covered by other values

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Process Type Type of process being studied.

- [ ] **Impact** Collision processes
- [ ] **Volcanic** Volcanic activity
- [ ] **Tectonic** Crustal deformation processes
- [ ] **Atmospheric** Atmospheric processes
- [ ] **Erosional** Surface degradation processes
- [ ] **Other** Any process type not covered by other values

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Composition Primary composition of the material being studied.

- [ ] **Silicate** Composed primarily of silicate minerals
- [ ] **Metal** Composed primarily of metallic elements
- [ ] **Organic** Composed primarily of organic compounds
- [ ] **Icy** Composed primarily of water or other volatile ices
- [ ] **Gaseous** Composed primarily of gas
- [ ] **Mixed** Composed of a mixture of different materials
- [ ] **Other** Any composition not covered by other values

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Environmental Context

Information about the environmental conditions at the study location.

Temperature Regime Temperature conditions at the study location.

- [ ] **Cryogenic** Extremely low temperatures, typically below 120K
- [ ] **Cold** Below freezing but not extreme
- [ ] **Temperate** Near-surface temperatures typical of Earth
- [ ] **Warm** Above freezing and below boiling point
- [ ] **Hot** Above boiling point
- [ ] **Other** Any temperature regime not covered by other values

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Pressure Conditions Pressure conditions at the study location.

- [ ] **Vacuum** Essentially no atmospheric pressure
- [ ] **Very Low** Minimal atmospheric pressure like Mars
- [ ] **Earth Like** Similar to Earth's atmospheric pressure
- [ ] **High** Higher than Earth's atmospheric pressure
- [ ] **Extreme** Very high pressure like Venus surface or gas giant interiors
- [ ] **Variable** Significant pressure variations or gradients
- [ ] **Other** Any pressure condition not covered by other values

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Radiation Environment Radiation conditions at the study location.

- [ ] **Low** Minimal radiation exposure like Earth surface
- [ ] **Moderate** Similar to Earth orbit or Mars surface
- [ ] **High** Significant radiation like Jupiter radiation belts
- [ ] **Extreme** Very high radiation levels
- [ ] **Shielded** Protected or shielded from external radiation
- [ ] **Other** Any radiation environment not covered by other values

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Atmospheric Presence Presence and nature of atmosphere at the study location.

- ☐ **None** No atmosphere present
- ☐ **Trace** Very low atmospheric pressure
- ☐ **Low** Atmospheric pressure similar to Earth's
- ☐ **High** Atmospheric pressure similar to Earth's
- ☐ **Extreme** Very high atmospheric pressure like Venus surface
- ☐ **Variable** Significant atmospheric pressure variations or gradients
- ☐ **Other** Any atmospheric presence not covered by other values

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Data Collection

Information about how the data was collected and its format.

Collection Method Method used to collect the data.

- ☐ **Orbiter** Data collected by an orbiting spacecraft
- ☐ **Lander** Data collected by a surface lander
- ☐ **Rover** Data collected by a mobile surface rover
- ☐ **Telescope** Data collected by Earth-based or space telescopes
- ☐ **Sample Return** Analysis of returned physical samples
- ☐ **Laboratory** Data collected in laboratory settings
- ☐ **Other** Any collection method not covered by other values

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Instrument Type Type of instrument used to collect the data.

- ☐ **Spectrometer** Instrument that measures spectra
- ☐ **Camera** Imaging device
- ☐ **Radar** Radio detection and ranging instrument
- ☐ **Lidar** Light detection and ranging instrument
- ☐ **Seismometer** Instrument that measures ground motion
- ☐ **Magnetometer** Instrument that measures magnetic fields
- ☐ **Other** Any instrument type not covered by other values

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Data Category General category of the collected data.

- ☐ **Altimetry** Elevation data
- ☐ **Image** Visual representation data
- ☐ **Photometry** Photometric measurement data
- ☐ **Map** Spatial representation data
- ☐ **Model Output** Results from computational models
- ☐ **Database** Flat rendering of existing data base
- ☐ **Parameter Data** Numerical data
- ☐ **Other** Any data category not covered by other values

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Data Format Format of the collected data.

- ☐ **Shape Model DTM** A 3D model of the surface of the object
- ☐ **Geologic Map** A map of the geologic features of the object
- ☐ **Thermal Map** A map of the thermal properties of the object
- ☐ **Mosaic** A composite image of the object
- ☐ **Spectral Map** A map of the spectral properties of the object
- ☐ **Gazetteer** A list of feature names
- ☐ **Time Series** Sequential data collected over time
- ☐ **Crater** A list of craters on the object
- ☐ **Light Curve** A list of light curves on the object
- ☐ **Spectral** A list of spectra on the object
- ☐ **Physical Properties** A list of physical properties of the object
- ☐ **Photometric Properties** A list of photometric properties of the object
- ☐ **Gravity Map** A map of the gravity of the object
- ☐ **Taxonomies** A list of taxonomies of the object
- ☐ **Crater Type** A list of irregular crater types on the object
- ☐ **Other** Any data format not covered by other values

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Temporal Epoch The temporal epoch of the data collection.

Earth Temporal epoch of the Earth

- ☐ **Eoarchean** Earth's 4,031 to 3,600 million years ago
- ☐ **Paleoarchean** Earth's 3,600 to 3,200 million years ago
- ☐ **Mesoarchean** Earth's 3,200 to 2,800 million years ago
- ☐ **Neoarchean** Earth's 2,800 to 2,500 million years ago
- ☐ **Paleoproterozoic** Earth's 2,500 to 1,600 million years ago
- ☐ **Mesoproterozoic** Earth's 1,600 to 1,000 million years ago
- ☐ **Neoproterozoic** Earth's 1,000 to 538.8 million years ago
- ☐ **Paleozoic** Earth's 238.8 to 251.9 million years ago
- ☐ **Mesozoic** Earth's 251.9 to 66 million years ago
- ☐ **Cenozoic** Earth's 66-0 million years ago
- ☐ **Ohter** Earth's Paleozoic Era

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Moon Temporal epoch of the Moon

- ☐ **Pre-Nectarian** Moon, 4,533 to 3,920 million years ago
- ☐ **Nectarian** Moon3,920 to 3,850 million years ago
- ☐ **Imbrian** Moon 3,850 to 3,200 millino years ago
- ☐ **Eratosthenian** Moon 3,200 to 1,100 million years ago
- ☐ **Copernican** Moon 1,100 to 0 million years ago

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Mars Temporal epoch of Mars

- ☐ **Pre-Noachian** Mars 4,500 to 4,100 million years ago
- ☐ **Noachian** Mars 4,100 to 3,700 million years ago
- ☐ **Hesperian** Mars 3,700 to 3,000 million years ago
- ☐ **Amazonian** Mars 3,000 to 0 million years ago

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Vesta Relative timescale for Vesta

- ☐ **Pre-Veneneia** \> 4,100 million years ago
- ☐ **Veneneia** ~ 4,100 million years ago
- ☐ **Rheasilvia** ~ 3,900 million years ago
- ☐ **Marcia** ~3,500 million years ago

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Ceres Relative timescale for Ceres

- ☐ **Pre-Kerwan** ~4,560 to 4,000 million years ago
- ☐ **Kerwan** 4,000 to 2,500 million years ago
- ☐ **Ernutet** ~ 2.500 to 0 million years ago

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Analysis Properties

Information about the analysis methods and properties of the data.

Software Used Software used to analyze the data.

- ☐ **Spectroscopy** Collection of spectra
- ☐ **Modeling** Computer-based simulation or modeling
- ☐ **Mapping** Collection of maps
- ☐ **Radar** Collection of radar data
- ☐ **Seismology** Collection of seismic data
- ☐ **Gravimetry** Collection of gravity data
- ☐ **Geochemistry** Collection of geochemical data
- ☐ **SEM** Collection of scanning electron microscopy data
- ☐ **TEM** Collection of transmission electron microscopy data
- ☐ **ICPMS** Collection of inductively coupled plasma mass spectrometry data
- ☐ **Other** Any software not covered by other values

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Research Context

Information about the research context and associated metadata.

Research Goals Research goals of the study

- [ ] **Accuracy** Accuracy of the data
- [ ] **Precision** Precision of the data
- [ ] **Other** Any research goal not covered by other values

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Related Publications Related publications of the study

Value: ______________________________

Keywords Keywords of the study

Value: ______________________________

Principal Investigator Principal investigator of the study

Value: ______________________________

Institution Institution of the study

Value: ______________________________
